

ARYAN BABURAJAN

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EXPERIENCE

Frappe.io Intern

Jan – Apr 2026

- Wrote a proof-of-concept Unicorn worker using Python 3.14 Subinterpreters, enabling true concurrent parallel execution beyond the limitations imposed by the GIL.
- Integrated Bloomberg's Memray profiler into Frappe's web dashboard to support memory and performance analysis.
- Extended Redis Queue's WorkerPool to allow workers to process multiple queues concurrently, improving memory usage by 32%.

PROJECTS

Ducktape Creator

2021 – 2025

- Built an open-source C++ 3D game engine with an Entity Component System architecture using EnTT, attaining 139+ GitHub stars and 28 forks.
- Implemented real-time rendering in OpenGL, with prior renderer implementations in Vulkan.
- Supports native C++ DLL-based scripting and an ImGui-based editor for in-engine debugging and content creation workflows.
- SerenityOS - Improved Unicode support across applications and migrated them to SerenityOS's GML markup language.
- NanoGPT - Implemented NanoGPT in PyTorch and extended it with Google's SentencePiece tokenizer for tokenization.
- Mirage - Product Hunt-style platform built with Next.js, React, MongoDB, Kinde Auth, and Resend for email newsletters.
- DirectShare - Built a peer-to-peer file transfer service with Express.js and EJS, deployed on Netlify.

ACHIEVEMENTS

1st Place (₹15K) - Google Developer Groups Physical AI Hackathon

Sep 2026

- Built a real-time telemetry collection and processing pipeline for Kerala's festive boat racing integrating an RPi with a cluster of ESP32s and flight controllers to give feedback on stroke sync and backwater detection.

Game Development Workshops Teacher

2025 - 2026
Thrissur, India

- Introduced an audience of 100+ to game development using Three.js.
- Mentored 50+ students in an introductory Unity game development workshop.
- Tutored 20+ students in a beginner game development session using p5.js.

TECHNICAL SKILLS

- **Languages:** C++, C, Rust, Python, TypeScript, JavaScript, C#
- **Web:** React, Next.js, Nest.js, Express.js, Node.js, WebSockets
- **Graphics:** OpenGL, Vulkan, EnTT, Dear ImGui, Unity, Blender
- **Tooling:** Git, CMake, Linux, MySQL, MongoDB, Netlify, Vercel, Heroku

EDUCATION

Christ College of Engineering
Bachelor of Technology in Computer Science and Engineering

2025 – Present
Kerala, India